Wei Wu

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SKILLS

Languages: Python, C++, C#, JavaScript, SQL, HTML/CSS, Bash
Backend & Web: FastAPI, Flask (Uvicorn/Gunicorn), Socket.IO, WebRTC (aiortc), REST APIs, asyncio (aiohttp/httpx), SQLModel/SQLAlchemy, Auth0 / OAuth
Frontend & Graphics: Vanilla-JS single-page apps, CanvasKit/Skia, WebAssembly (Emscripten/Rust), Vue.js, Bootstrap 5
Computer Vision: OpenCV, NumPy, SciPy, scikit-image, Shapely, trimesh/Open3D, ChArUco/homography, sub-pixel detection, dithering
AI / ML: PyTorch, CUDA, TensorRT, HuggingFace Transformers, Segment-Anything (SAM2), Depth-Anything-V2, BiRefNet
Data & Cloud: MySQL, SQLite, Redis, Azure (IoT Hub, Blob Storage, Key Vault)
DevOps & Infra: Docker, Portainer, Gitea, TeamCity, Traefik, nginx, systemd, FRP, Paramiko, Selenium, PyInstaller
Embedded & Real-time: Raspberry Pi CM4/CM5, RP2040/PIO firmware, galvo control, Picamera2, pyserial/I2C/GPIO

EDUCATION

Carnegie Mellon University May 2023 | GPA 3.93/4.0
M.S., Mechanical Engineering & Engineering Technology Innovation Management
Coursework: Computer Vision (OpenCV), Engineering Computation (C++), Linear Control (Python)

University of Wisconsin–Madison May 2021 | GPA 3.72/4.0
B.S., Mechanical Engineering; Certificate in Entrepreneurship (Dean's List, 6 semesters)

EXPERIENCE

Full Spectrum Laser — Las Vegas, NV · *Full Stack Software Engineer (Backend Lead)* July 2023 – Present

- Built and solely own the **async task-queue offload service** (Flask/Gunicorn/multiprocessing) — workers pull tasks from a shared queue and execute each as a sequential multi-stage pipeline, with multiple tasks running in parallel across the pool; added per-stage timing logs for observability; diagnosed and fixed a **race-condition deadlock** on shared state under concurrent dequeues; hardened against OOM respawn storms with live memory monitoring.
- Drove the RE4 platform backend migration from **synchronous Flask to async FastAPI + Socket.IO** — new codebase from scratch: modular event handlers, a server-wide blocking→async refactor (39 files), MySQL/SQLModel data layer, and OAuth (Auth0) login; **serves 500+ users**.
- Designed and owned the design-to-machine-instructions compiler backend that turns user design files (SVG, DXF, PDF, images, 3D point clouds) into the compact binary instruction format the device firmware executes — binary writer/reader, geometry transforms, a preview renderer, a job-time estimator, and config models for 20+ machine variants — including a **Traveling-Salesman toolpath optimizer** (Python + parallel C++ with kd-tree) that **cut engraving time 72%** for 7% added processing.
- Built a **multi-server in-memory state cache** (asyncio-locked, race-condition-free) in front of MySQL: conditional write-through, fetch-on-miss fallback, offline-authoritative mode, and cross-server invalidation over HTTP — caching device security keys to sign jobs without a per-request cloud round-trip.
- Designed a **headless Jobs REST API** (submit / stop / status) with **content-hash caching** to avoid re-uploading identical files and one-time-password (TOTP) device authentication, for fully scriptable job control.
- Independently built and deployed **image.fslaser.com (~1,000 conversions/month)** — Flask + Bootstrap web app **backed by GPU AI model servers**: SAM2, Depth-Anything-V2, BiRefNet background removal, and Florence-2 natural-language selection.
- Owned **containerization & CI/CD** for the server fleet, moving the team off TeamCity to a self-hosted Docker / Gitea-registry / Portainer pipeline: authored every Dockerfile and buildx script for the web platform (production/beta/alpha), the offload worker pool, and three GPU AI model servers; provisioned bare-metal GPU servers (NVIDIA/CUDA, FRP tunnels, SSH hardening) and air-gapped offline devices, plus a nightly Selenium visual-regression suite.
- Designed and shipped the **over-the-air (Azure IoT) fleet-update system** — A/B release slots for safe rollback, a cloud device-state watcher, production-device safety gating, and an automated board-swap migration tool, enabling fleet-wide updates with no SSH — and built the low-latency browser-to-device control channel (WebRTC, with Socket.IO/server fallback and auto-reconnect).
- Solely built and shipped a **$2,500 offline-server appliance** — hardened headless Ubuntu (LUKS + TPM2 auto-unlock, Secure Boot), re-architected the web stack so every cloud dependency runs locally, and configured the box as a router + WiFi AP (dnsmasq, hostapd, NAT, nginx); shipped per-customer units to B2B clients.
- **Principal front-end author of RE5** (re5.fslaser.com), the current-generation in-browser design editor — a vanilla-JS single-page app rendering with **Skia/CanvasKit** and offloading heavy work to WebAssembly: shortest-path toolpath optimization, image rasterization & dithering, font-to-outline text, auto-nesting of shapes to save material, level-of-detail rendering, and a parametric-CAD mode backed by a 2D geometric-constraint solver.
- Built an in-browser (vanilla-JS) 3D point-cloud engine for engraving designs inside glass/crystal, moving 4 of 6 generation paths off the GPU server into the browser: a **from-scratch 16-bit PNG decoder** (zero dependencies), a compact binary codec, depth-aware layered sampling, and an in-place Three.js depth editor that cut per-edit latency from a **1–3-second server round-trip to instant**.
- Architected the laser-scanner calibration & computer-vision pipeline — OpenCV detection of printed calibration markers (ChArUco) and grids with **sub-pixel accuracy**, curve fitting, and a robust RANSAC homography that maps 225 grid points from hardware units to real-world millimeters — plus stitching many overlapping camera frames into one image (alignment, flat-field correction, denoising); compiled the detector to WebAssembly.
- Optimized **on-device motion-control streaming** — replaced a per-point Python generator with a vectorized NumPy producer pipeline to eliminate buffer underruns on dense curves; added an adaptive settle-delay model (delay scales with turn sharpness); built live camera streaming over WebRTC for real-time preview.
- Earlier (2023): built RE3, the **C# / Blazor** engraving app that runs on the machine's built-in screen — adding 3D engraving inside glass/crystal — plus the Flask AI backend service.

MathWorks — Natick, MA · *Autonomous Driving Product Marketing Intern* May – Aug 2022

- Ran competitive analysis of RoadRunner vs. rival ADAS simulation tools; findings directly shaped the 2-year product roadmap.
- Relayed customer feature requests to engineering; gained familiarity with the ADAS simulation stack (sensor fusion, scenario generation, HD maps).

PROJECTS

- **ADAS Semantic Segmentation** — *Carnegie Mellon University, 2022* — Improved a VGG16-UNet model from 73% → 86% accuracy with shadow-mitigation pre-processing and OpenCV contour-correction post-processing.
- **UAV Flight Simulator** — *Carnegie Mellon University, 2021* — Built a drone flight simulator in C++/OpenGL with OOP-modeled aerodynamics, automatic obstacle avoidance, and a real-time interactive GUI. github.com/afiretony/24780-Engineers-Republic